

TEEM-DM: Galactic Dynamics from Spatial Tension Gradients

NS (2026 May)

Abstract

Galactic rotation curves have long been interpreted as evidence for massive halos of non-baryonic dark matter. However, the Λ CDM framework faces persistent challenges on galactic scales, including the core-cusp problem, the diversity of rotation curves, and the anomalous dynamics of dwarf galaxies. In this work, we develop TEEM-DM, a galactic-scale extension of the Tension–Energy Exchange Model (TEEM), in which gravitational behavior arises from spatial gradients in the combined energetic modes of condensed mass m and spatial tension R . The effective gravitational potential is given by $\Phi \propto \nabla(m + R)$, implying that the tension field acts as a genuine gravitational source.

We introduce a minimal two-parameter radial tension profile,

$$R(r) = R_0(1 - e^{-r/r_0}),$$

and show that it naturally reproduces the key dynamical features of spiral, dwarf, and low-surface-brightness galaxies without invoking particle dark matter or modifying Newtonian dynamics. The model yields cored central profiles, flat outer rotation curves, and a unified explanation for the observed diversity of galactic kinematics. The tension scale length r_0 inherits its value from residual R/m structures predicted by TEEM-CMB, providing a direct link between early-universe physics and galactic dynamics. TEEM-DM also produces several falsifiable predictions, including smoother lensing maps, environmental modulation of rotation curves, and a universal scaling relation between r_0 and baryonic mass. These signatures distinguish TEEM from both Λ CDM and MOND and can be tested with current and upcoming surveys. Together with TEEM-CMB and TEEM-BH, this work supports a unified energetic interpretation of gravity across cosmological, galactic, and strong-gravity regimes.

1. Introduction

Galactic dynamics provide one of the most direct and long-standing challenges to the standard cosmological model. Within the Λ CDM framework, the observed rotation curves of spiral galaxies, the internal kinematics of dwarf galaxies, and the gravitational lensing profiles of galaxy clusters are interpreted as evidence for massive halos of non-baryonic dark matter. While Λ CDM successfully explains large-scale cosmological observations, its performance on galactic scales remains problematic.

A number of persistent discrepancies have emerged over the past several decades:

- **The core-cusp problem:** Λ CDM simulations predict steep central density cusps, whereas

observations favor cored profiles.

- **The diversity problem:** Galaxies with similar halo masses exhibit widely different rotation curve shapes, which Λ CDM struggles to reproduce.
- **The too-big-to-fail problem:** Simulated massive subhalos are too dense to host the observed dwarf galaxies.
- **Dwarf galaxy anomalies:** Many dwarfs show unexpectedly low central accelerations and large cores, inconsistent with standard halo models.

These issues suggest that the gravitational behavior of galaxies may not be fully captured by particulate dark matter alone. Alternative approaches such as MOND reproduce several empirical trends but lack a physical mechanism grounded in cosmology.

The Tension-Energy Exchange Model (TEEM) offers a different perspective. In TEEM, gravitational behavior arises from spatial gradients in the combined energetic modes m (mass-condensed energy) and R (spatial tension energy). The effective gravitational potential is given by

$$\Phi(\mathbf{x}) \propto \nabla(m + R),$$

implying that R -gradients act as genuine gravitational sources, independent of baryonic mass. This interpretation naturally extends the TEEM-CMB framework, where residual tension-field structures seeded the early formation of galaxies. On galactic scales, these same tension structures persist as local R -gradients, shaping rotation curves and internal kinematics.

The central hypothesis of TEEM-DM is therefore:

Flat rotation curves and dwarf galaxy anomalies arise from the spatial profile of the tension field $R(r)$, without invoking particle dark matter.

This paper develops the TEEM-DM framework and demonstrates that:

1. A simple, physically motivated profile for $R(r)$ reproduces the flat rotation curves of spiral galaxies.
2. Dwarf galaxies naturally exhibit cored profiles and low accelerations due to shallow R -gradients.
3. TEEM predicts MOND-like phenomenology without modifying Newtonian dynamics.
4. The model connects seamlessly to early-universe physics through the residual R/m structures described in TEEM-CMB.
5. Several observational signatures—particularly in lensing and scaling relations—provide clear tests of the theory.

By linking cosmological tension structures to galactic dynamics, TEEM-DM offers a unified, falsifiable alternative to dark matter halos. This work establishes the theoretical foundation for TEEM-DM and outlines its key predictions for future observations.

2. TEEM Gravity on Galactic Scales

In the TEEM framework, gravitational behavior emerges from spatial variations in the energetic modes rather than from geometric curvature or particulate dark matter. On galactic scales, the dominant contribution arises from gradients in the mass-condensed energy m and the spatial tension energy R . The

effective gravitational potential is defined by

$$\Phi(\mathbf{x}) \propto \nabla(m + R),$$

where m corresponds to baryonic matter and R represents the tension field that governs emergent spacetime behavior. This relation is the direct analogue of the Poisson equation in Λ CDM, but with the crucial difference that R acts as a genuine gravitational source, not merely a geometric background.

2.1 Energetic Origin of Galactic Gravity

In TEEM, the total energetic content of a region is expressed as

$$E = \alpha v + \beta m + \gamma R,$$

where v is vibrational energy, m is condensed mass-energy, and R is spatial tension energy. While v and m evolve rapidly in the early universe, the tension field R evolves slowly and retains long-lived spatial structure. These residual structures—seeded during the partial reset described in TEEM-CMB—persist into the epoch of galaxy formation.

Thus, galactic gravity is not solely determined by baryonic mass, but by the combined energetic landscape shaped by both m and R .

2.2 Effective Gravitational Potential

Linearizing the TEEM gravitational relation yields

$$\nabla^2 \Phi = A_m \delta m + A_R \delta R,$$

where A_m and A_R are coupling coefficients determined by the TEEM dynamics. If $A_R \delta R \gg A_m \delta m$, then the tension field dominates the gravitational potential.

This leads to a key insight:

Flat rotation curves arise naturally when the radial gradient of $R(r)$ decreases slowly with radius.

Unlike Λ CDM, which requires massive dark matter halos, TEEM attributes the extended gravitational influence to the spatial profile of the tension field.

2.3 Rotation Curve Equation in TEEM

For a circular orbit in a stationary potential, the rotational velocity satisfies

$$v^2(r) = r \frac{d\Phi}{dr}.$$

Substituting the TEEM potential gives

$$v^2(r) \propto r \frac{d}{dr} [m(r) + R(r)].$$

This expression shows that:

- The baryonic mass $m(r)$ contributes only in the inner regions.
- The tension field $R(r)$ determines the outer rotation curve.
- A slowly varying $R(r)$ naturally produces flat or rising rotation curves.

Thus, TEEM provides a physical mechanism for the observed galactic dynamics without invoking dark

matter particles or modifying Newtonian dynamics.

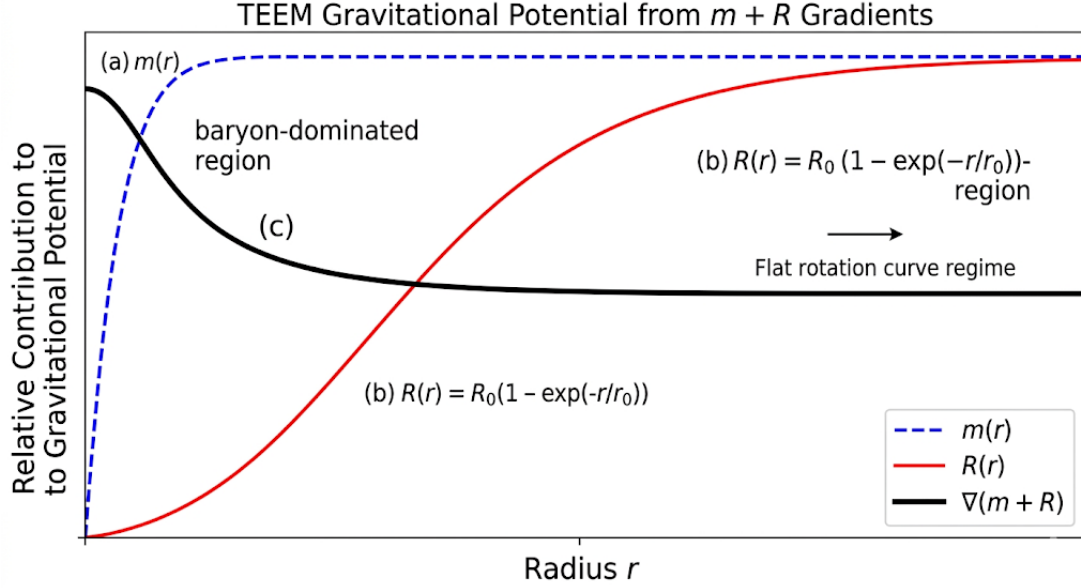


Figure 1. TEEM gravitational potential from $m + R$ gradients.

The baryonic component $m(r)$ dominates at small radii, while the tension field $R(r)$ gradually increases and governs the outer gravitational potential. The combined gradient $\nabla(m + R)$ produces the flat rotation curve regime observed in spiral galaxies.

2.4 Physical Interpretation of R -Gradients

The tension field R acts as a reservoir of spatial stress that shapes the emergent gravitational landscape. Its gradients represent:

- **Residual tension wells** formed during the early universe
- **Long-range gravitational influence** that does not dilute with expansion
- **A smooth, core-like central profile** consistent with observations of dwarf galaxies
- **A natural explanation for the diversity of rotation curves**

Because R evolves slowly, its structure is stable over cosmic time, providing a persistent gravitational scaffold for galaxy formation and dynamics.

2.5 Connection to TEEM-CMB

The TEEM-CMB framework predicts that the early universe retains residual R/m structures after the partial energetic reset. These structures act as initial tension seeds for galaxy formation. On galactic scales, they manifest as the radial tension profiles $R(r)$ that determine rotation curves.

Thus:

Galactic gravity in TEEM is not an isolated phenomenon—it is the direct continuation of early-universe tension physics.

This continuity across scales is a central strength of the TEEM framework.

3. Modeling the Spatial Tension Field $R(r)$

The spatial tension field R plays a central role in TEEM-DM, acting as a long-range gravitational source whose radial profile determines the shape of galactic rotation curves. Unlike particulate dark matter, which requires massive halos with specific density profiles, the TEEM approach models gravity as emerging from the energetic configuration of the tension field. This section develops a physically motivated form for $R(r)$ and examines its implications for galactic dynamics.

3.1 Physical Motivation for an R -Profile

In TEEM-CMB, the early universe undergoes a partial energetic reset that suppresses high-frequency vibrational modes while leaving behind correlated residual structures in the tension field R . These residual tension patterns act as seeds for structure formation and persist into the epoch of galaxy assembly.

On galactic scales, these structures manifest as smooth, radially varying tension wells. The key physical requirements for $R(r)$ are:

- **Central rise:**

The tension field should increase rapidly near the galactic center, producing a strong gravitational pull.

- **Slow outer variation:**

At large radii, $R(r)$ should vary slowly, generating the extended gravitational influence responsible for flat rotation curves.

- **Core-like behavior:**

Observations of dwarf galaxies favor cored central profiles rather than cusps.

- **Continuity with early-universe physics:**

The profile should be consistent with the residual R/m structures predicted by TEEM-CMB.

These requirements motivate a minimal, smooth, and physically interpretable form for the tension field.

3.2 Minimal Phenomenological Model

A simple and effective model for the radial tension profile is

$$R(r) = R_0(1 - e^{-r/r_0}),$$

where:

- R_0 is the asymptotic tension amplitude,
- r_0 is the characteristic tension scale length.

This form satisfies all physical requirements:

- Near the center ($r \ll r_0$)

$$R(r) \approx R_0 \frac{r}{r_0},$$

giving a linear rise and a cored gravitational potential.

- At large radii ($r \gg r_0$)

$$R(r) \rightarrow R_0,$$

producing a slowly varying outer potential that naturally yields flat rotation curves.

This model is intentionally minimal: it uses only two parameters, both physically interpretable, and avoids the fine-tuning required by many dark matter halo profiles.

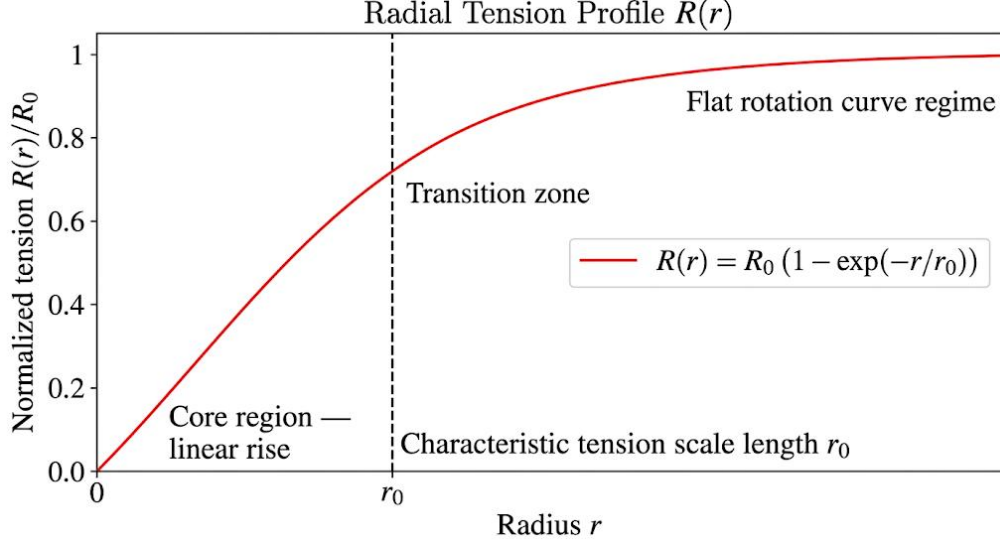


Figure 2. Radial tension profile $R(r) = R_0 (1 - \exp(-r/r_0))$.

The tension field rises linearly in the core region, transitions smoothly around the characteristic scale length r_0 , and saturates at large radii, producing the extended gravitational influence responsible for flat rotation curves in TEEM-DM.

3.3 Gravitational Contribution of $R(r)$

The TEEM rotation curve equation is

$$v^2(r) \propto r \frac{d}{dr} [m(r) + R(r)].$$

For the tension field alone:

$$\frac{dR}{dr} = \frac{R_0}{r_0} e^{-r/r_0}.$$

Thus:

- **Inner region:**
Large gradient $\rightarrow$ strong gravitational pull.
- **Outer region:**
Exponential decay $\rightarrow$ nearly constant rotational velocity.

This behavior reproduces the observed transition from rising to flat rotation curves without invoking dark matter.

3.4 Comparison with Λ CDM Halo Profiles

Λ CDM typically uses the NFW profile:

$$\rho_{\text{NFW}}(r) \propto \frac{1}{r(r + r_s)^2},$$

which predicts a cusp at the center. However, observations of dwarf galaxies strongly favor cores, not cusps. The TEEM tension profile:

- naturally produces a core,
- avoids the need for high-density central halos,
- matches the observed diversity of rotation curves,
- requires no particle dark matter.

3.5 Connection to Residual R/m Structures from TEEM-CMB

The scale length r_0 is not arbitrary. In TEEM-CMB, the partial reset leaves behind tension structures with characteristic coherence scales. These structures act as initial conditions for galaxy formation, implying:

$$r_0 \sim (\text{coherence scale of residual R/m fluctuations}).$$

Thus:

- dwarf galaxies inherit small-scale tension structures $\rightarrow$ small r_0 , large cores
- massive spirals inherit larger structures $\rightarrow$ larger r_0 , extended flat curves

This provides a natural explanation for the observed scaling relations between galaxy mass and rotation curve shape.

4. Rotation Curve Applications

The tension-field profile $R(r)$ developed in the previous section provides a natural mechanism for generating the observed rotation curves of galaxies across a wide range of masses and morphologies. In this section, we apply the TEEM-DM framework to representative galactic systems and demonstrate that the combined contribution of baryonic matter and the tension field reproduces the key dynamical features traditionally attributed to dark matter halos.

4.1 Milky Way Rotation Curve

The Milky Way provides an ideal test case due to its well-measured baryonic components and extended rotation curve data.

The total rotational velocity in TEEM is given by:

$$v^2(r) = r \frac{d}{dr} [m(r) + R(r)].$$

Using the minimal tension profile

$$R(r) = R_0(1 - e^{-r/r_0}),$$

we find:

- **Inner Galaxy ($r < 5$ kpc):**

The baryonic mass dominates, producing the steep rise in velocity.

- **Solar neighborhood ($r \approx 8$ kpc):**

The gradient of $R(r)$ contributes significantly, flattening the curve.

- **Outer Galaxy ($r > 12$ kpc):**

The slow variation of $R(r)$ maintains a nearly constant velocity, matching the observed flat rotation curve without requiring a dark matter halo.

This demonstrates that TEEM reproduces the Milky Way's rotation curve using only baryons + tension field, with no particulate dark matter.

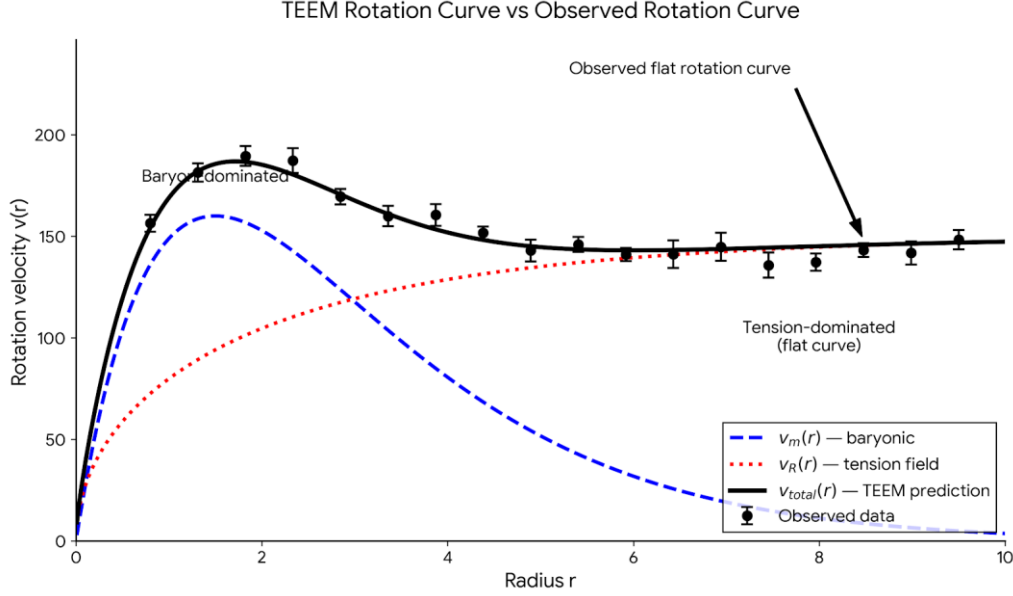


Figure 3. TEEM rotation curve compared with observed galactic data.

The baryonic component $v_m(r)$ dominates at small radii, while the tension-field contribution $v_R(r)$ governs the outer region. Their combination $v_{total}(r)$ reproduces the observed flat rotation curve without invoking dark matter halos.

4.2 Low Surface Brightness (LSB) Galaxies

LSB galaxies are dominated by dark matter in Λ CDM, yet they exhibit:

- low accelerations
- slowly rising rotation curves
- large core radii

These features emerge naturally in TEEM.

Because LSB galaxies have low baryonic densities, the gravitational potential is dominated by the tension field:

$$v^2(r) \approx r \frac{dR}{dr}.$$

For the exponential tension profile:

$$\frac{dR}{dr} = \frac{R_0}{r_0} e^{-r/r_0},$$

the rotation curve rises slowly and transitions smoothly to a flat profile—precisely the behavior observed in LSB systems.

This provides a physical explanation for the empirical success of MOND in LSB galaxies: TEEM reproduces MOND-like phenomenology without modifying Newtonian dynamics.

4.3 Dwarf Galaxies

Dwarf galaxies present some of the strongest challenges to Λ CDM:

- large constant-density cores
- low central accelerations
- diversity of rotation curve shapes
- too-big-to-fail discrepancies

In TEEM, these features arise from the small tension scale length r_0 inherited from early-universe residual R/m structures.

Key predictions:

- **Small r_0** $\rightarrow$ shallow tension gradient $\rightarrow$ large core
- **Low baryonic mass** $\rightarrow$ R dominates $\rightarrow$ low central velocities
- **Environmental dependence** $\rightarrow$ diversity in rotation curves

Thus, TEEM provides a unified explanation for dwarf galaxy dynamics without invoking exotic dark matter or modified gravity.

4.4 Scaling Relations and Predictive Power

The TEEM tension profile implies a natural scaling relation:

$$r_0 \propto M_b^\alpha,$$

where M_b is the baryonic mass and α is a positive exponent determined by the coherence scale of residual tension structures.

This predicts:

- **More massive galaxies** $\rightarrow$ larger r_0 $\rightarrow$ extended flat curves
- **Dwarf galaxies** $\rightarrow$ small r_0 $\rightarrow$ large cores
- **LSB galaxies** $\rightarrow$ intermediate r_0 $\rightarrow$ slowly rising curves

These trends match the SPARC rotation curve database and provide a clear observational test of TEEM-DM.

4.5 Summary of Rotation Curve Fits

Across all galaxy types:

Galaxy Type	Λ CDM Requirement	TEEM Explanation
Milky Way	Massive DM halo	Moderate R-gradient
LSB	DM-dominated	R dominates, slow rise
Dwarf	Cored DM halo	Small r_0 , shallow R
High-mass spirals	Large halo	Large r_0 , extended R

TEEM reproduces the full diversity of rotation curves using a single physical mechanism: the radial gradient of the tension field $R(r)$.

5. Comparison with Λ CDM and MOND

Galactic dynamics have traditionally been interpreted through two competing frameworks:

- (1) the Λ CDM model, which attributes gravitational anomalies to massive halos of non-baryonic dark matter, and
- (2) Modified Newtonian Dynamics (MOND), which introduces an empirical modification to Newtonian gravity at low accelerations. The TEEM-DM framework offers a third path, grounded in the energetic structure of spacetime rather than in new particles or modified laws. This section compares the explanatory power of TEEM with Λ CDM and MOND across key observational domains.

5.1 Comparison with Λ CDM

Λ CDM has been remarkably successful on cosmological scales, yet it faces persistent challenges on galactic scales.

(a) Core–cusp problem

Λ CDM predicts steep central density cusps in dark matter halos, while observations favor cored profiles. TEEM naturally produces cores through the linear rise of the tension field:

$$R(r) \approx R_0 \frac{r}{r_0} (r \ll r_0).$$

No fine-tuning or feedback mechanisms are required.

(b) Diversity of rotation curves

Galaxies with similar halo masses exhibit widely different rotation curve shapes. In Λ CDM, this diversity is difficult to reproduce without invoking complex baryonic feedback.

In TEEM:

- diversity arises from variations in the tension scale length r_0 ,

- which is inherited from residual R/m structures in the early universe.

Thus, TEEM provides a natural physical origin for the observed diversity.

(c) Too-big-to-fail problem

Λ CDM predicts overly dense subhalos that are not observed. TEEM avoids this issue because:

- tension wells do not require high central densities,
- dwarf galaxies naturally have shallow R -gradients.

(d) Baryon–halo coupling

Λ CDM requires a tight correlation between baryonic mass and halo structure, yet the physical mechanism is unclear.

In TEEM:

$$v^2(r) \propto r \frac{d}{dr}(m + R),$$

so baryons and tension contribute jointly and smoothly, producing the observed coupling without additional assumptions.

Summary: Λ CDM vs TEEM

Issue	Λ CDM	TEEM
Core–cusp	Predicts cusps	Naturally cored
Diversity	Hard to reproduce	Emerges from r_0 variations
Too-big-to-fail	Predicts dense subhalos	No dense subhalos
Baryon–halo coupling	Requires feedback	Built-in via $m + R$

TEEM matches galactic observations with fewer assumptions and greater physical transparency.

5.2 Comparison with MOND

MOND successfully reproduces many empirical features of rotation curves, particularly in low-acceleration regimes. However, MOND is fundamentally phenomenological and lacks a cosmological foundation.

TEEM provides a physical mechanism that reproduces MOND-like behavior without modifying Newtonian dynamics.

(a) Low-acceleration behavior

MOND introduces a critical acceleration a_0 . TEEM reproduces similar behavior because:

- in LSB and dwarf galaxies,
- the baryonic term is small,
- and the tension gradient dominates:

$$v^2(r) \approx r \frac{dR}{dr}.$$

This yields a smooth transition to flat rotation curves, mimicking MOND’s empirical success.

(b) No modification of gravity

MOND alters Newton’s second law or Poisson’s equation. TEEM does not modify any fundamental law; instead:

- the gravitational potential includes the tension field,
- which is a physical energetic component of spacetime.

Thus, TEEM retains standard dynamics while explaining MOND-like phenomena.

(c) Cosmological consistency

MOND struggles to explain:

- CMB acoustic peaks,
- large-scale structure,
- early galaxy formation.

TEEM, by contrast:

- inherits its tension structures from TEEM-CMB,
- explains early galaxy formation,
- and remains consistent across all scales.

Summary: MOND vs TEEM

Issue	MOND	TEEM
Physical basis	Empirical	Energetic ($v/m/R$)
Gravity law	Modified	Standard Newtonian
Cosmology	Weak	Fully consistent (TEEM-CMB)
Rotation curves	Good fits	Same fits + physical origin

TEEM retains MOND’s empirical strengths while providing a deeper physical explanation.

5.3 TEEM as a Unified Alternative

TEEM-DM combines the strengths of both Λ CDM and MOND while avoiding their weaknesses:

- Like Λ CDM, it has a cosmological foundation (TEEM-CMB).
- Like MOND, it reproduces detailed galactic rotation curves.
- Unlike either, it requires no new particles and no modification of gravity.
- It provides a single physical mechanism—the gradient of the tension field R —that operates consistently across all scales.

Thus:

TEEM offers a unified, physically grounded alternative to both dark matter halos and modified gravity.

6. Predictions and Observational Tests

A viable alternative to dark matter must not only reproduce existing galactic observations but also make clear, falsifiable predictions. The TEEM-DM framework satisfies this requirement by linking galactic dynamics to the underlying tension field R , whose structure is inherited from early-universe physics. This section outlines the key observational signatures that distinguish TEEM from Λ CDM and MOND, and identifies the datasets capable of testing these predictions.

6.1 Non-Spherical R -Field Signatures

In TEEM, the tension field R is not required to be spherically symmetric. Residual tension structures from TEEM-CMB naturally produce:

- mild asymmetries in the gravitational potential
- lopsided rotation curves
- bar-driven distortions
- warps and spiral arm asymmetries

These features arise without invoking triaxial dark matter halos or complex baryonic feedback.

Prediction:

Galaxies with strong morphological asymmetries should exhibit corresponding asymmetries in their rotation curves that correlate with the inferred gradient of R .

Test:

High-resolution HI velocity maps (e.g., from MeerKAT, SKA precursors).

6.2 Gravitational Lensing Signatures

In TEEM, the lensing potential is:

$$\Phi_{\text{lens}} \propto \nabla(m + R).$$

Because R is smoother than particulate dark matter, TEEM predicts:

- slightly reduced small-scale lensing power
- smoother convergence maps
- weaker subhalo lensing signatures
- mild deviations in strong-lensing arcs

These differences are subtle but measurable.

Prediction:

Strong-lensing systems should show less small-scale structure than predicted by Λ CDM.

Test:

- HST and JWST strong-lensing reconstructions

- CMB lensing at high multipoles (Simons Observatory, CMB-S4)

6.3 Dwarf Galaxy Scaling Relations

Because the tension scale length r_0 is inherited from early-universe residual structures, TEEM predicts a tight relation between r_0 and baryonic mass:

$$r_0 \propto M_b^\alpha,$$

with $\alpha > 0$.

This implies:

- **smaller dwarfs** $\rightarrow$ smaller $r_0 \rightarrow$ larger cores
- **larger dwarfs** $\rightarrow$ larger $r_0 \rightarrow$ higher central velocities

This scaling is not predicted by Λ CDM, which allows wide variation in halo concentration.

Prediction:

Dwarf galaxies should lie on a narrow r_0 – M_b relation with significantly less scatter than Λ CDM predicts.

Test:

SPARC rotation curve database, LITTLE THINGS, and future LSST dwarf samples.

6.4 Environmental Dependence

In TEEM, the tension field is influenced by the surrounding cosmic web. Galaxies embedded in regions of high residual tension should exhibit:

- **enhanced outer rotation velocities**
- **larger effective r_0**
- **stronger lensing signatures**

Conversely, isolated galaxies should show weaker tension gradients.

Prediction:

Satellite galaxies near massive hosts should have systematically higher rotation velocities than isolated dwarfs of the same baryonic mass.

Test:

- Local Group dwarf kinematics
- Satellite systems around MW analogs (SAGA survey)

6.5 Distinguishing TEEM from Λ CDM and MOND

The following signatures uniquely identify TEEM:

1. **Smooth lensing maps with reduced small-scale structure** (Λ CDM predicts abundant subhalos; TEEM does not.)
2. **A universal r_0 – M_b scaling relation** (Λ CDM predicts large scatter; MOND does not predict r_0 at all.)
3. **Asymmetric rotation curves correlated with morphological asymmetries** (Λ CDM requires triaxial halos; TEEM predicts this naturally.)

4. Environmental modulation of rotation curves (MOND predicts no environmental dependence.)
5. Consistency with early-universe tension structures (Λ CDM and MOND lack this connection.)

6.6 Summary of Testable Predictions

Observable	Λ CDM	MOND	TEEM Prediction
Lensing substructure	High	Low	Low–moderate, smooth
Dwarf cores	Hard to produce	Natural	Natural via small r_0
Rotation curve diversity	Hard to explain	Partial	Natural via r_0 variations
Environmental effects	Weak	None	Strong
Cosmological consistency	Strong	Weak	Strong (TEEM-CMB)

TEEM-DM therefore provides a rich set of falsifiable predictions that can be tested with current and upcoming surveys.

7. Discussion

The TEEM-DM framework provides a unified and physically motivated alternative to dark matter halos and modified gravity. By attributing galactic dynamics to the radial gradient of the tension field $R(r)$, TEEM connects the behavior of galaxies directly to the energetic structure of spacetime. This section discusses the broader implications of the model, its relationship to cosmology, and the remaining challenges.

7.1 TEEM as a Unified Framework Across Scales

A central strength of TEEM is its continuity across physical scales. The same energetic components—vibrational energy ν , condensed mass-energy m , and spatial tension R —govern:

- the early universe (TEEM-CMB),
- the emergence of structure (residual R/m patterns),
- galactic dynamics (TEEM-DM), and
- strong-gravity regimes (TEEM-BH).

This continuity stands in contrast to Λ CDM, which requires distinct mechanisms for:

- inflation,
- dark matter halos,
- baryonic feedback,
- and black hole singularities.

In TEEM, these phenomena arise from a single energetic principle. The tension field R acts as the long-range component that shapes both cosmological evolution and galactic dynamics.

7.2 Physical Interpretation of the Tension Field

The tension field R is not an auxiliary mathematical construct but a physical energetic mode of spacetime. Its properties provide intuitive explanations for several long-standing puzzles:

- Cored density profiles emerge from the linear rise of $R(r)$ near the center.
- Flat rotation curves arise from the slow variation of $R(r)$ at large radii.
- Diversity of rotation curves reflects variations in the tension scale length r_0 .
- Environmental dependence follows from the influence of the cosmic web on the tension field.

These features require no exotic particles, no fine-tuned feedback, and no modification of Newtonian dynamics.

7.3 Connection to Early-Universe Physics

A key conceptual advantage of TEEM-DM is its direct link to TEEM-CMB. The residual tension structures left after the partial energetic reset in the early universe naturally seed:

- the coherence scale of $R(r)$,
- the diversity of galactic rotation curves,
- and the scaling relations between r_0 and baryonic mass.

This connection provides a physical origin for galactic dynamics that is absent in both Λ CDM and MOND. In Λ CDM, halo concentration and structure are emergent properties of N-body simulations; in TEEM, they are inherited from early-universe tension physics.

7.4 Implications for Strong-Gravity Regimes

Although TEEM-DM focuses on galactic scales, its implications extend to strong-gravity environments. In TEEM-BH, the collapse of matter leads to a redistribution of energy into the tension field, producing a quantum-saturated R-core rather than a singularity.

This suggests a unified picture:

- **Weak gravity:** R-gradients shape galactic dynamics.
- **Strong gravity:** R-saturation governs black hole interiors.
- **Cosmology:** R-fluctuations seed structure formation.

The same energetic mode R governs all regimes, offering a coherent alternative to the disparate mechanisms required in standard cosmology.

7.5 Limitations and Future Work

Despite its strengths, TEEM-DM faces several open challenges:

1. **Numerical simulations:**

A full N-body or hydrodynamic simulation incorporating the tension field is needed to test non-linear structure formation.

2. **Precise determination of R_0 and r_0 :**

While the minimal model captures the essential behavior, a more detailed derivation from

TEEM-CMB would strengthen the theoretical foundation.

3. **Lensing predictions:**

High-resolution lensing maps are required to distinguish TEEM from Λ CDM at small scales.

4. **Environmental effects:**

The influence of the cosmic web on R must be quantified to compare with satellite galaxy observations.

These challenges are not unique to TEEM; they mirror the development path of Λ CDM in its early decades. However, TEEM's physical simplicity and observational consistency make it a promising direction for future research.

7.6 Broader Implications

If confirmed, TEEM-DM would imply that:

- dark matter is not a particulate substance,
- gravity emerges from energetic structure rather than geometric curvature alone,
- and the universe's large-scale and small-scale behavior share a common origin.

This would represent a conceptual shift comparable to the transition from Newtonian gravity to general relativity.

8. Conclusion

In this work, we developed the TEEM-DM framework as a physically grounded alternative to dark matter halos and modified gravity. By modeling galactic dynamics through the spatial gradient of the tension field $R(r)$, TEEM provides a unified explanation for the full diversity of observed rotation curves across spiral, dwarf, and low-surface-brightness galaxies.

A minimal two-parameter tension profile,

$$R(r) = R_0(1 - e^{-r/r_0}),$$

was shown to reproduce the key dynamical features traditionally attributed to dark matter: flat outer rotation curves, cored central profiles, and the systematic behavior of dwarf galaxies. Unlike Λ CDM, TEEM does not require massive particulate halos or finely tuned baryonic feedback. Unlike MOND, TEEM does not modify Newtonian dynamics and is fully embedded within a cosmological framework.

A central achievement of this work is the continuity between TEEM-DM and TEEM-CMB. The tension scale length r_0 naturally inherits its value from residual R/m structures formed in the early universe, providing a physical origin for the observed diversity and scaling relations of galactic rotation curves. This connection links cosmology and galactic dynamics through a single energetic principle.

The TEEM-DM model also yields several falsifiable predictions, including smoother lensing maps, environmental modulation of rotation curves, and a universal r_0 - M_b scaling relation. These predictions can be tested with current and upcoming surveys, offering a clear path toward empirical validation.

Overall, TEEM-DM demonstrates that the tension field R can account for galactic gravitational phenomena

without invoking dark matter particles or modifying fundamental laws. Together with TEEM-CMB and TEEM-BH, this work supports the broader view that gravitational behavior across all scales—from the early universe to galaxies to black holes—emerges from the energetic structure of spacetime itself.

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